

Philippe FONTAINE

Data Scientist - Economist

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Summary

Data scientist at the French Senate (previously at the General Inspectorate of Finance and the French Treasury). Specialised in fraud detection, public policy evaluation and the analysis of large-scale datasets (SNDS, tax data, national accounts). Over 10 years of experience in quantitative studies, impact assessments and the development of decision-support tools (R, Python, Spark, Shiny, cloud).

Professional experience

2025 – **Data Scientist**, French Senate, Paris

- **Evaluation work:** quantitative support for the evaluation and oversight of government action (data analysis, impact assessments, drafting of briefing notes).
- **Legislative work:** quantitative analyses, simulations and summaries on economic, social and budgetary topics.

2021 – 2025 **Data Scientist**, General Inspectorate of Finance (IGF), Paris

- **Fraud detection** (energy transition, international trade): developed a rule engine and a reference database containing several million prices from open sources (*web scraping*); built a *no-code* application enabling business teams to reproduce the analyses autonomously. *R, Shiny*
- **Evaluation of the national healthcare strategy:** used the SNDS and DREES databases to analyse several billion healthcare claims and simulate reforms of the long-term condition scheme (ALD). *Spark, R, microsimulation, public finances*
- **Decision-support tools:** developed two Shiny applications (HR management; mission tracking and GDPR compliance) and an interactive dashboard of central government revenues (Eurostat data) used by the Budget Directorate as part of the 2024 spending reviews. *R, Shiny, Python*
- **Public policy evaluation:** several economic studies (tobacconists, nursing homes, financial sector; see publications).

2018 – 2021 **Deputy Head of Unit FIPU2**, *FIPU Subdirector*ate, French Treasury, Paris

- Corporate tax analysis and costing; major contribution to assessments and negotiations on the introduction of a global minimum tax promoted by the OECD ([BEPS project](#)).
- Modernised and automated the production of general government accounts.

2015 – 2018 **Head of Investment and Tax**, *Quarterly Accounts Department*, Insee, Paris

- Oversaw the quarterly estimation of investment and taxes using econometric models and short-term indicators.
- Produced tax revenue forecasts for Insee's economic outlook publication.

Technical skills

Programming & data	R (Shiny, Quarto, data.table), Python (pandas, scikit-learn), SQL, Spark, Git
Web	HTML, CSS, JavaScript, Bootstrap 5
Statistics / ML	Econometrics, predictive modelling, time series, microsimulation
Economics	Public finances, national accounting, macroeconomics

Infrastructure / Application containerisation (Docker), deployment and CI/CD pipelines (Git-DevOps Lab), security and GDPR, large-scale data processing (Spark, distributed databases)

Education

- 2019 **Master's degree (M2) – Public Policy Evaluation and Decision-Making**, ENSAI, Rennes (35)
French National School for Statistics and Information Analysis
- 2013 – 2015 **Insee statistician officer programme**, ENSAI, Rennes (35)
- 2011 – 2013 **MPSI/MP preparatory classes (maths and physics)**, *Lycée Alphonse Daudet*, Nîmes (30)

Publications

- 2025 [Additional construction costs linked to regulatory standards](#)
- 2025 [Enforcing sentences: 20 proposals to improve enforcement and give meaning to criminal sanctions](#)
- 2025 [Transparency and evaluation of state aid to businesses: a democratic demand and a key to economic efficiency](#)
- 2025 [Changes in the financing of efforts to combat violence against women](#)
- 2025 [Mental health and psychiatry: no 'great national cause' without substantial resources](#)
- 2025 [25 proposals to give municipal police the tools to tackle everyday insecurity](#)
- 2025 [Employment of older workers: a key to safeguarding the pay-as-you-go pension system?](#)
- 2025 [Tackling the high cost of living in the overseas territories: sticking plasters or real remedies?](#)
- 2024 [Spending review on long-term conditions: towards a more efficient and equitable scheme](#)
- 2023 [Territorial allocation of the compensation for the CVAE \(value-added contribution of businesses\)](#)
- 2023 [Cross-cutting methodological mission on the State's fields of intervention and public policy instruments](#)
- 2022 [Schooling of pupils with disabilities](#)
- 2022 [Mission on the management of nursing homes \(EHPAD\) in the Orpea group](#)
- 2021 [Evaluation committee for the France Relance recovery plan](#)
- 2021 [Evaluation committee for the reforms of capital taxation](#)
- 2020 [New rules for the international taxation of corporations: assessment of the economic effects for France](#)

Full list available at www.philippefontaine.eu

Interests

Cybersecurity (TryHackMe, HackTheBox), web application development and open-source projects

Defence, geopolitical issues and social psychology

Electronic music production